INCH-POUND

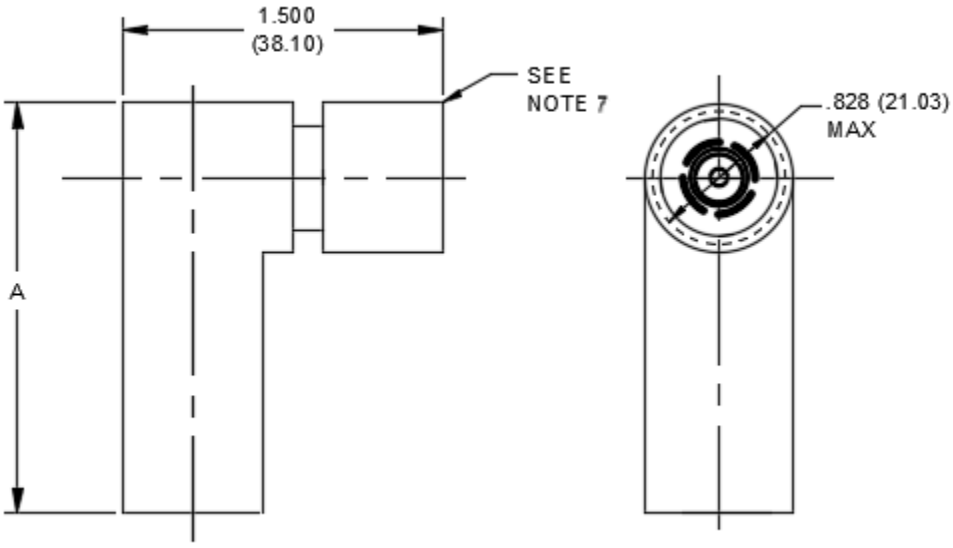
MIL-PRF-39012/39H
w/AMENDMENT 2
12 October 2021
SUPERSEDING
MIL-PRF-39012/39H
w/AMENDMENT 1
21 November 2016

PERFORMANCE SPECIFICATION SHEET

# CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY (SERIES SC (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.




NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. For dimension A, see tables I and III.
3. Dimension A is the maximum envelope dimension.
4. Wrench flats to accommodate standard wrench in accordance with FED-STD-H28.
5. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Series SC, pin contacts, in accordance with MIL-STD-348.
8. Three holes .027 (0.69 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nut is optional.

FIGURE 1. General configuration.

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TABLE I. Dash numbers, cross-reference, and dimensions.

| Dash no. 1/ | Applicable cable M17/ 2/ | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| CATEGORY A – FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 3/ | | | |
| 0016<br>Cable group VIII | 2-RG6 4/<br>180-00001 4/<br>73-RG212<br>162-00001<br>188-00001<br>199-00001<br>112-RG304 5/<br>171-00001 | A | 1.875 (47.63) |
| 0101<br>Cable group X | 6-RG11 4/<br>181-00001 4/<br>62-RG144 4/<br>65-RG165<br>159-00001<br>74-RG213<br>189-00001<br>163-00001<br>75-RG214<br>190-00001<br>164-00001<br>86-00001 6/<br>127-RG393 5/<br>174-00001<br>77-RG216<br>191-00001 | | |
| 0018<br>Cable group XII | 78-RG217 5/<br>165-00001<br>188-00001 | | |
| 0019<br>Cable group XIII | 72-RG211<br>161-00001 | | |
| 0020<br>Cable group XIV | 79-RG218 5/<br>166-00001<br>193-00001 | | |
| 0106<br>Cable group XI | 74-RG215 5/<br>189-00002 | | |
| 0107<br>Cable group IX | 92-RG115 5/ 6/<br>168-00001 | | |

See footnotes at end of table.



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TABLE I. Dash numbers, cross-reference, and dimensions. – Continued.

| Dash no. 1/ | Applicable cable M17/ 2/ | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| CATEGORY C – FIELD REPLACEABLE (SAE AS22520 CRIMP TOOL) SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE 3/ 7/ | | | |
| 0010<br>Cable group VIII | 2-RG6 4/ 8/<br>180-00001 4/ 8/<br>73-RG212 8/<br>162-00001 8/<br>188-00001 8/<br>199-00001 8/<br>112-RG304 5/ 8/<br>171-00001 8/ | A | 2.125 (53.98) |
| 0004<br>Cable group XA | 65-RG165 5/ 6/ 9/<br>159-00001 9<br>74-RG213 9/<br>189-00001 9/<br>163-00001 9/ | | |
| 0005<br>Cable group XB | 75-RG214 9/<br>190-00001 9/<br>164-00001 9/<br>86-00001 6/ 9/<br>127-RG393 5/ 9/<br>174-00001 9/ | | |
| 0013<br>Cable group XC | 6-RG11 4/ 9/<br>181-00001 4/ 9/<br>62-RG144 4/ 5/ 6/ 9/ | | |
| 0009<br>Cable group IX | 92-RG115 5/ 6/ 9/<br>168-00001 9/ | | |
| CATEGORY D – FIELD REPLACEABLE – DEFINED PIECE PART 3/ 7/ 10/ | | | |
| 0501<br>Cable group XB | 75-RG214<br>190-00001<br>164-00001<br>86-00001 6/<br>127-RG393 5/<br>174-00001 | A | 2.125 (53.98) |
| 0502<br>Cable group XA | 65-RG165 5/<br>159-00001<br>74-RG213<br>189-00001<br>163-00001 | | |
| 0503<br>Cable group VIB | 60-RG142 6/<br>158-00001<br>84-RG223<br>167-00001<br>194-00001<br>200-00001<br>128-RG400 5/<br>175-00001 | A | 2.125 (53.98) |

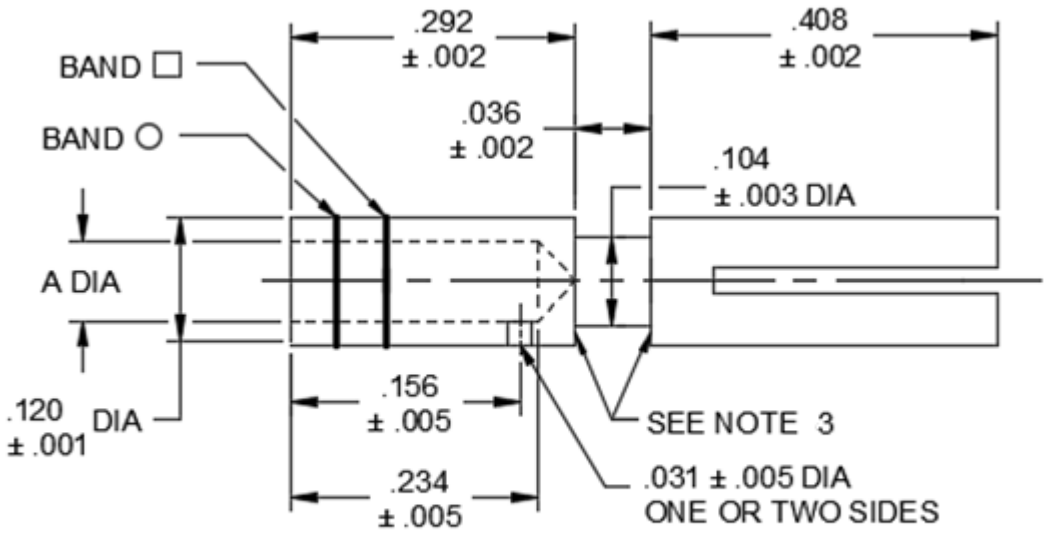
See footnotes at end of table.



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TABLE I. Dash numbers, cross-reference, and dimensions – Continued.

1/ For cross-reference of dash number to superseded PIN or type designation, see table IV.
2/ The latest version of each cable shall be applicable.
3/ These connectors have captivated center contacts.
4/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.
5/ Cable to be used when performing tests requiring cable except as in 4/ and 6/.
6/ Cable to be used for the 200°C temperature cycling tests.
7/ These connectors are assembled, using the applicable crimp tool, to the specified cable stripper as shown on figure 3.
8/ M22520/5-35 closure A or M22520/5-55 closure A.
9/ M22520/5-61.
10/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.




CENTER CONTACT

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .043 | 1.09 | .250 | 6.35 | .418 | 10.62 |
| .002 | 0.05 | .098 | 2.49 | .292 | 7.42 | .438 | 11.13 |
| .003 | 0.08 | .104 | 2.64 | .351 | 8.92 | .492 | 12.50 |
| .005 | 0.13 | .120 | 3.05 | .375 | 9.52 | .500 | 12.70 |
| .015 | 0.38 | .156 | 3.96 | .394 | 10.01 | .600 | 15.24 |
| .031 | 0.79 | .220 | 5.59 | .406 | 10.31 | | |
| .036 | 0.97 | .234 | 5.94 | .408 | 10.36 | | |

FIGURE 2. Contact and ferrule dimensions for category D only.

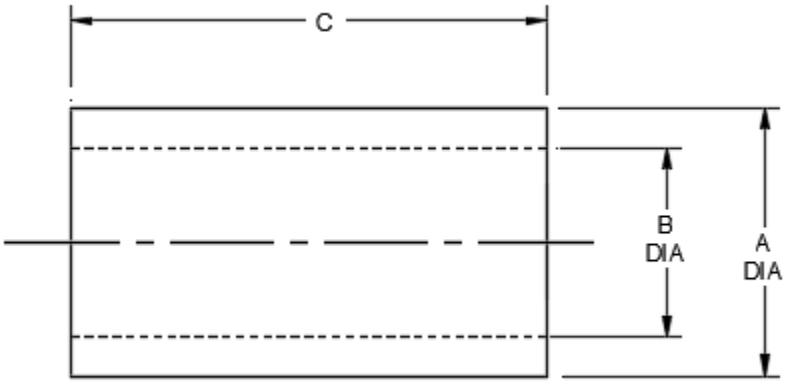


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| Dash no | Contact no 1/ | A (inches) | Basic crimp tool 2/ | Crimp die or positioner | Crimp tensile minimum | Color band | Color Band O |
|---|---|---|---|---|---|---|---|
| 0501<br>0502 | 39-10 | .098 ± .002 | M22520/1-01 | M22520/1-13 | 60 pounds | Red | Orange |
| 0503 | 39-11 | .043<br>+.001<br>-.002 | | | 20 pounds | Blue | |

1/ Contacts numbers are for identification only.
2/ Class 2 tool may be used by OEM (see SAE AS22520).

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. Crimp tensile test shall be in accordance with SAE AS39029.
3. Maximum break of .003 inch (0.08 mm).
4. Copyright notice: All information disclosed in these specification sheet which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. Form socket to mate with 063/066 diameter pin.
6. Color bands shall be positioned such that no coloring material enters the inspection hole.



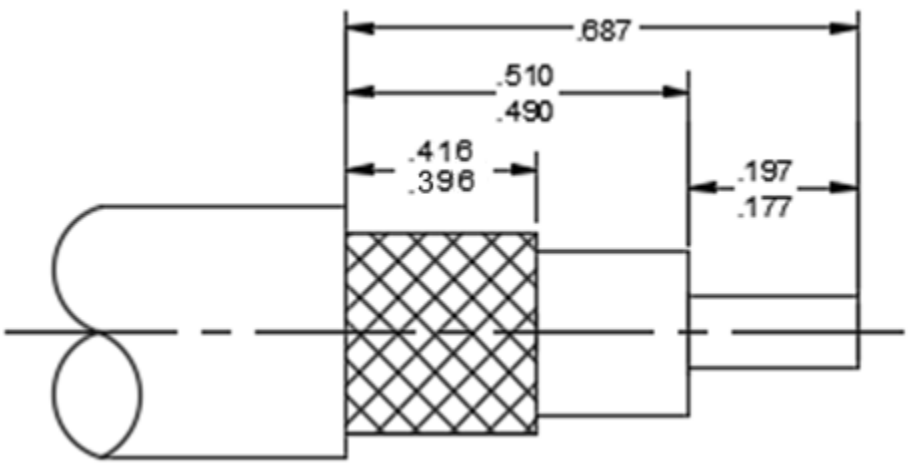

| Dash number | Ferrule number 1/ | A | B | C ± .015 | Basic crimp tool 2/ | Crimp die or positioner M22520/5- |
|---|---|---|---|---|---|---|
| 0501 | 39-50 | .492 ± .003 | .438 ± .003 | .600 | M22520/5-01 | 25 Closure A or 61 |
| 0502 | 39-51 | .492 ± .003 | .418 ± .003 | .600 | | |
| 0503 | 39-52 | .250 ± .003 | .220 ± .003 | .500 | | 5, 11, 57 Closure A or 19 Closure B |

1/ Ferrule numbers are for identification only.
2/ Class 2 tools may be used by OEM (see SAE AS22520).

FIGURE 2. Contact and ferrule dimensions for category D only – Continued.



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| Inches | mm |
|---|---|
| .177 | 4.50 |
| .197 | 5.00 |
| .396 | 10.06 |
| .416 | 10.57 |
| .490 | 12.45 |
| .510 | 12.95 |
| .687 | 17.45 |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.

FIGURE 3. Cable stripping dimensions for field replaceable connectors.

ENGINEERING DATA:

Nominal impedance: 50 ohms

Frequency range: 0 to 11,000 MHz

Voltage rating

1,000 volts rms maximum working voltage at sea level.

250 volts rms maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIREMENTS:

Design and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds.

Inspection conditions: Coupling torque: 6-10 inch pounds.

Mating characteristics: See MIL-STD-348 for dimensions.

Outer contact:

Test ring ID: .411 maximum, 16 microinch finish.

Insertion force: 15 pounds maximum when inserted a minimum of .125.

Contacts with slotted members: Shall contact a .419 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.



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Leakage (pressurized connectors) Not applicable.

Insulation resistance: MIL-STD-202-302, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only) 15 pounds minimum axial force for all cables except RG400 and RG142, 6 pounds minimum for RG400 and RG142.

Corrosion (salt spray): MIL-STD-202-101, test condition B.

Voltage standing wave ratio (VSWR) From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whoever is lower.

From 500 to 9,000 MHz 1.35 maximum

From 9,000 to 11,000 MHz 1.50 maximum

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.015 + .005 F (F in GHz).

Item 16: VSWR shall be less than 1.015 + .005 F (f in GHz).

Second step of VSWR checkout procedure – VSWR shall be less than 1.045 + .019 F (F in GHz)

Group B inspection – VSWR shall be less than 1.10 + .01 F (F in GHz)

Qualification and group C inspection – VSWR shall not exceed 1.15.

Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

| | Initial | After environment |
|---|---|---|
| Center contact | 2.0 | 2.5 |
| Outer contact | .15 | Not applicable |
| Braid to body | .05 | Not applicable |

Dielectric withstanding voltage. MIL-STD-202-301. 3,000 volts rms minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms minimum for connectors using RG400 and RG142.

Vibration, high frequency. MIL-STD-202-204, test condition B.

Shock: MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: MIL-STD-202-106. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 750 volts rms minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 2,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.



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Cable retention force:

Noncrimp assemblies: 75 pounds minimum

Crimp assemblies.

50 lbs minimum for cables 155-189 OD

60 lbs minimum for cables 190-.229 OD

75 lbs minimum for cables 230-249 OD

90 lbs minimum for cables 250 OD and larger

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz. This requirement may be meet by qualifying the straight connector version for the same cable under MIL-PRF-39012/35.

Insertion loss:

.21 dB maximum at 9 GHz.

.07 $\sqrt{F}$ (GHz) dB max tested at 3 GHz and 6 GHz.

Part or Identifying Number (PIN): M39012/39- (dash number from table I or "B" number from table III).

TABLE II. Group qualification. 1/

| Group | Submission and qualification of any of the following connectors 2/ M39012/39 | Qualifies the following connectors M39012/39 |
|---|---|---|
| I | -0101 | -0101<br>-0106<br>-0107 |
| II | B0002 | B0002<br>B0003<br>B0008 |
| III | -0004 | -0004<br>-0005<br>-0009 |
| IV | -0501<br>-0502 | -0501<br>-0502<br>-0503 |
| V | -0503 | -0503 |

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within the same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right-hand column. The part does not necessarily have to be the part initially qualified.



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TABLE III. CATEGORY B – nonfield replaceable (special tools may be required). 1/

| NOT FOR ARMY, NAVY, OR AIR FORCE USE. FOR OEM USE ONLY |
|---|

| PIN 2/ 3/ M39012/39B | Applicable cable 4/ M17/ | Dimensions | Inches (millimeters) Maximum |
|---|---|---|---|
| 0002 | 65-RG165 5/<br>74-RG213 | A | 2.125 (53.98) |
| 0003 | 75-RG214 6/<br>RG-225/U 5/ | | |
| 0008 | 92-RG115 5/<br>112-RG304 | | |

1/ For maintenance replacements for category B, see table V.
2/ For cross-reference of dash number to superseded PIN or type designation, see table IV.
3/ These connectors have captivated center contacts.
4/ The latest version of each cable shall be applicable.
5/ Cable to be used for the +200°C temperature cycling tests.
6/ Cable to be used when performing tests requiring cable except as in 5/.

TABLE IV. Supersession data. 1/

| Preferred PIN<br><br>M39012/39 | Superseded PIN or type designation M39012/39 |
|---|---|
| -0101 | -0001 |
| -0106 | -0006 |
| -0107 | -0007 |
| B002 | -0002 |
| B003 | -0003 |
| B008 | -0008 |

1/ The superseded PIN or the type designation is for reference only. Where a superseded PIN or type designation is not given, none was assigned or will be assigned. The PIN M39012/39-XXXX shall be used in all cases for marking and identifying the connector.

TABLE V. Maintenance replacements for category B.

| Category B number 1/ | Category C dash number | Category A dash number | Category D dash number |
|---|---|---|---|
| B0002 | 0004 | 0101 | 0502 |
| B0003 | 0005 | 0101 | 0502 |
| B0008 | 0009 | 0107 | 0501 |

1/ Category B connectors are for original installation only. They will not be stocked or procured by the Government.



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Referenced documents. In addition to MIL-PRF-39012, this document references the following;

FED-STD-H28
MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-202-301
MIL-STD-202-302
MIL-PRF-39012/35
MIL-STD-348
SAE AS22520
SAE AS39029

CONCLUDING MATERIAL

Custodian:
Army – CR
Navy – EC
Air Force – 85
NASA – NA
DLA - CC

Preparing activity:
DLA – CC

Review activities:
Army – AT, AV, EA
Navy – AS, MC, OS
Air Force – 19

(Project 5935-2021-116)

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